

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Assumptions, Limitations, Risks & Dependencies Report

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance
and Healthcare Support Using Data and AI
Track: Data Science

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Week 7 Submission – Consolidated Review of Weeks 4–7

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This report is a Week 7 Data Science track deliverable for the HealthConnect Clinic Experience Lab engagement. It consolidates the assumptions, limitations, risks and dependencies identified across the track's Week 4 Risk and Dependency Register, the Week 5 Review of Week 4 Foundation, the Week 5 Project Execution & Progress Package, the Week 6 Model Improvement & Validation Report, and the Week 7 Model Testing & Refinement Report (the eight-test T01–T08 testing cycle and the subsequent refinement and retesting work), presenting a single, current picture of what has been resolved, what remains open, and what has newly emerged, ahead of the Week 8 Final Presentation.

Maintaining a consolidated, actively reviewed register of this kind is standard project management practice, consistent with the Project Management Institute's guidance on identifying, analysing and responding to project risk throughout a project's life cycle (Project Management Institute [PMI], 2021), and with the NIST AI Risk Management Framework's guidance that AI-specific risks — including bias, reliability and safety — should be tracked and managed continuously rather than assessed once (Tabassi, 2023). This report should be read alongside, rather than instead of, the five source documents named above; it does not repeat their full technical content, only the items relevant to assumptions, limitations, risks and dependencies.

Key Assumptions Carried Forward

The following assumptions, first stated in Week 4, remain in effect and continue to shape how findings from this project should be interpreted:

- The HealthConnect Appointment Dataset is fictional and synthetic; all findings are illustrative of methodology rather than clinically validated, and this limitation is restated explicitly in every report produced by this track.
- The model is intended to support clinic staff in prioritising outreach for high-risk appointments, not to automate scheduling or intervention decisions.
- Only information genuinely available at the time of booking should be used as a model input; this assumption underpins the exclusion of `waiting_time_minutes` and the leakage-safe design of the Week 6 threshold-optimisation work.
- HealthConnect has not yet supplied a formal cost weighting between a missed no-show (False Negative) and an unnecessary patient contact (False Positive); until it does, this track assumes that reducing missed appointments should not come at a disproportionate cost to the patient support experience, consistent with the Week 4 problem framing.
- Patient identifiers, while not present in the current processed feature matrices, are assumed to be recoverable from the original dataset if required for patient-grouped validation; this remained unresolved through Week 7's own testing cycle (T01, T06) and is carried forward to Week 8 (see Sections 3 and 4).
- The complete, 38-feature Week 6 candidate specification (three engineered interaction features plus both calendar features) is the correct basis for all Week 8 integration and deployment work, following the Week 7 calendar-feature ablation's rejection under its pre-declared tolerance (see Section 1).

1. Issues that have been resolved

The following issues, first identified in Weeks 4 and 5, have been substantively resolved by the end of Week 6:

Issue	Originally Identified	Resolution
Repeated-patient leakage risk — 5,000 appointments across only 1,696 patients risked patient overlap between train and test sets under a standard random split.	Week 5	Resolved: a patient-aware GroupShuffleSplit (grouped by patient_id) was adopted, with zero patient overlap between the training and test sets confirmed and carried forward unchanged into Week 6.
waiting_time_minutes leakage ambiguity (R1) — unclear whether the field was available at booking time or only after the appointment.	Week 4	Mitigated: excluded from the feature set as a precaution; a Week 5 t-test found no significant group difference ($p = 0.781$), supporting the decision. Formal data-owner confirmation is still outstanding (Section 2).
Missing threshold-sensitivity evidence — an earlier version of the Week 6 report flagged that narrative references to threshold_df / threshold_df_refined tables could not be verified against an executed notebook cell.	Week 6	Resolved: the revised Week 6 notebook constructed leakage-safe threshold_df_refined and threshold_business_df tables from out-of-fold cross-validated probabilities, and this evidence is now fully reported in the Week 6 Model Improvement & Validation Report, Section 6.3.
Feature-refinement hypotheses left unvalidated — three interaction features engineered from the error analysis had not yet been confirmed to improve performance without overfitting.	Week 6	Resolved: the Refined Logistic Regression was retrained on the refined feature set and compared fairly against the unmodified baseline, confirming a modest but genuine ROC-AUC/PR-AUC gain with no material increase in overfitting (Week 6 report, Sections 5 and 5.1).
Ambiguity between the Week 4 proposed chronological split and an alternative patient-aware split.	Week 4 → 5	Resolved: the chronological split was deliberately superseded by a patient-aware grouped split, documented as an explicit, reasoned decision rather than left ambiguous (Week 5 Project Summary, Section 7).

The following issues, first identified in Week 6 or earlier, were substantively resolved during Week 7 testing (T01–T08, reported in full in the Week 7 Model Testing & Refinement Report):

Issue	Originally Identified	Resolution
Feature-engineering gain stability unconfirmed (NI3) — the Week 6 ROC-AUC/PR-AUC improvement over	Week 6	Resolved (tested, and found not robust): T01's five-fold cross-validation shows ROC-AUC

Issue	Originally Identified	Resolution
baseline had not been tested for robustness.		ranging 0.6521–0.6970 (not a fixed figure), and T02's fold-matched comparison shows the candidate winning only 3 of 5 folds on ROC-AUC and 2 of 5 on PR-AUC, with slightly negative mean differences (-0.0008 ROC-AUC, -0.0006 PR-AUC). The open question is now answered, even though unfavourably; this is carried forward as a standing interpretive limitation (Section 8) rather than an open unknown.
Segment-level performance untested — Week 6 had not evaluated whether model errors were concentrated in identifiable patient or appointment groups.	Week 6	Resolved: T04 evaluated 40 segment/group combinations and identified a clear lead-time trade-off (short-lead-time bookings under-flagged, long-lead-time bookings over-flagged) and a smaller distance-based effect (Week 7 report, Section 6).
Mandatory cross-track requirements not end-to-end validated — Week 6's Data Analytics exchange (D9) had not been followed by a joint retest of shared findings.	Week 6	Resolved: the mandatory T08 cross-track activity harmonised Data Science's segment definitions to the Data Analytics Week 7 lead-time bands and prior-no-show-history groups, and passed on both tested directional findings (Week 7 report, Section 13).
Calendar-feature coefficient-variability concern — two calendar-derived features showed higher coefficient variability than the engineered interaction features, raising an unresolved refinement question.	Week 7 (T05)	Resolved (decided): tested via a pre-declared ± 0.002 tolerance ablation; once a feature-reconstruction gap (see new issue NI5 below) was corrected, the ablation was rejected, and the complete 38-feature specification was retained as the Week 7 refined model (Week 7 report, Section 8).

2. Issues that remain unresolved

The following issues, first identified in Weeks 4, 5 or 6, remain open as of the end of Week 7 and require action before or during Week 8:

Issue	Originally Identified	Current Status
Cancellation-exclusion definition (R2) — excluding Cancelled appointments	Week 4	Open; not yet confirmed with HealthConnect stakeholders.

Issue	Originally Identified	Current Status
from the target definition may not match the clinic's actual operational priorities.		
Subgroup fairness untested (R3) — model performance across demographic and access-related subgroups has not been formally assessed.	Week 4	Open; Week 7's T04 segment testing (Week 7 report, Section 6) provided descriptive error-rate breakdowns by gender, age group and other segments, but a formal disparate-impact or fairness test (for example, demographic parity or equal opportunity) has still not been performed on the Refined Logistic Regression or its provisional 0.45 threshold.
Sunday-appointment anomaly unescalated (R7) — 737 appointments (14.7%) are scheduled on Sundays despite the Clinic Knowledge Base stating the clinic is closed that day.	Week 5	Open; still not escalated to the data owner or mentor as of Week 7. <code>is_sunday_appointment</code> continues to be used as a live model feature and was tested descriptively as a Week 7 T04 segment (Week 7 report, Section 6), but the anomaly's underlying cause remains unconfirmed.
README/notebook split-method description (R8, T14) — the README's description of the train/test split method has not been confirmed as updated to reflect the patient-aware method actually used.	Week 5	Open; the substantive metrics-reporting inconsistency was resolved in Week 5, but this specific documentation item has still not been confirmed as corrected as of Week 7.
<code>waiting_time_minutes</code> formal confirmation (R1, D2) — the exclusion decision is well supported internally but has not been formally confirmed by the data owner.	Week 4	Open; internally mitigated, externally unconfirmed.
Cross-track syncs with Machine Learning Engineering, Project Management and Generative AI (R6) — proposed in the Week 5 Track Coordination Plan.	Week 4	Partially open; the Week 7 T08 mandatory cross-track activity adds a second, more rigorous Data Analytics exchange (Section 4, D10) on top of the Week 6 dashboard artefact (D9). The Machine Learning Engineering, Project Management and Generative AI syncs have still not been confirmed as completed.
Provisional operating threshold not yet stakeholder-confirmed — the HealthConnect-aligned 0.45 threshold was selected using training-only	Week 6	Open; Week 7's T07 (Week 7 report, Section 3.2) quantified the trade-off precisely (186 false negatives avoided for 187

Issue	Originally Identified	Current Status
evidence, not a HealthConnect-supplied cost weighting.		additional false positives, approximately 1.005 extra false positives per false negative avoided) but no formal HealthConnect stakeholder tolerance for that trade-off has yet been obtained.
Categorical variables not formally tested for association with no-show status — gender, appointment_type and reminder_sent have not been tested with a method such as a chi-square test of independence.	Week 5	Open; Week 7's T04 segment testing (Week 7 report, Section 6) evaluates these variables descriptively but does not substitute for a formal independence test.
Prediction-level appointment-ID traceability — the Data Analytics track could not trace individual model predictions back to a specific appointment_id in the evidence supplied for the T08 cross-track activity.	Week 7	Open; explicitly not claimed as resolved by the T08 cross-track activity (Week 7 report, Section 13).
Missing-distance data discrepancy — Data Analytics' source data shows 86 eligible records with missing distance data, versus 39 in the Data Science working dataframe used for T08.	Week 7	Open; not yet investigated by this track (Week 7 report, Section 13).
ML Engineering pipeline validation — confirming that the saved model artefact, feature schema and threshold are consumed correctly by any downstream serving pipeline.	Week 6 (D6)	Open; now specifically targets the retained 38-feature specification rather than the previously-reported 35-feature ablated one (Section 4, D12).

3. New issues discovered

The following issues were newly discovered during Week 6 and had not been documented in the Week 4 or Week 5 registers:

ID	Issue	Description
NI1	Absence of patient identifiers in processed feature matrices	The processed Week 5/6 feature matrices (X_train_refined, X_test_refined) do not retain patient_id, so the Week 6 threshold-selection cross-validation could only use ordinary stratified k-fold rather than the patient-grouped separation used for the original train/test split — reintroducing a milder form of the patient-leakage risk that Week 5 had otherwise resolved for model training.
NI2	Untuned tree-based models overfit substantially	Random Forest (train ROC-AUC 0.9903 vs. test 0.6637) and Histogram Gradient Boosting (train

ID	Issue	Description
		ROC-AUC 0.8766 vs. test 0.6615) both showed severe train–test performance gaps with default hyperparameters, meaning neither can currently be treated as a reliable alternative to Logistic Regression.
NI3	Feature-engineering gain is small and not yet proven stable	The Refined Logistic Regression's improvement over the Week 5 baseline (+0.0017 ROC-AUC, +0.0037 PR-AUC) is small enough that it has not yet been demonstrated to be robust to the specific train/test split, rather than a sample-specific artefact.
NI4	No formal cost weighting supplied by HealthConnect	Threshold selection had to proceed using an internally defined intervention-efficiency rule (Extra False Positives per False Negative Avoided) rather than a HealthConnect-confirmed relative cost of a missed appointment versus an unnecessary patient contact — a dependency not previously documented as a risk.

The following issues were newly discovered during Week 7 testing and had not been documented in any earlier register:

ID	Issue	Description
NI5	Feature-reconstruction gap (resolved within Week 7)	The Week 7 testing notebook's helper function originally rebuilt only two of the Week 6 candidate's three engineered interaction features, omitting history_interaction, so early Week 7 testing ran on a 37-feature reconstruction rather than the full 38-feature Week 6 specification. Corrected during the week; all figures in the final Week 7 report reflect the complete 38-feature reconstruction.
NI6	T02 narrative/executed-output mismatch (resolved within Week 7)	An interpretation cell stated a 1-in-5 fold win rate for the candidate against the Week 5 baseline, where the test's own executed comparison table shows 3-in-5. Corrected within the week.
NI7	Standard-deviation computation bug (resolved within Week 7)	The T05 feature-review table computed its Std column over six values (five fold coefficients plus an already-appended Mean) instead of five, understating true coefficient variability. The notebook's calculation and both of its dependent interpretation cells were corrected.
NI8	Recurring narrative/executed-output staleness pattern (resolved within Week 7)	A series of narrative interpretation cells — covering T01's and T03's confusion-matrix/error-summary text, the final-holdout interpretation cell's ROC-AUC figure, the auto-generated README's key-metrics section, and the 'Testing Findings and Issues Identified' summary cell's T02, T03, T04 and T05 subsections — were left

ID	Issue	Description
		unregenerated after the 38-feature retrain (NI5). Identified and corrected across several iterative verification passes; every figure in the final Week 7 report has been independently checked against the notebook cell that produced it.
NI9	Appointment-type label mismatch in T04 narrative (open)	A T04 narrative commentary cell cites an "appointment_type" group labelled "New Patient" that does not appear among the segment's four actual executed groups (Diagnostic Test, Follow-up, General Consultation, Specialist Consultation). The Week 7 report relies only on the executed segment tables, but the mismatch itself has not yet been corrected in the notebook.
NI10	Calendar-feature ablation decided on a narrow margin (open)	The ablation was rejected on a Recall@0.45 change of -0.0021 against a pre-declared ± 0.002 tolerance; a different, equally defensible tolerance value could plausibly have produced the opposite (ACCEPT) decision. This reproducibility-oriented sensitivity was not apparent until NI5 was corrected and the ablation re-run on the complete feature set.
NI11	Elevated coefficient variability in retained calendar features (open)	The retained appointment_day_of_year and appointment_month features carry materially higher coefficient variability across folds than the three Week 6 interaction features; this should be monitored if the model is retrained on new data.

4. New cross-track dependencies

Building on the Week 4 dependency register (D1–D5), the following cross-track dependencies were newly identified, or newly evidenced, during Week 6:

ID	Dependency	Track(s)	Status
D6	The Machine Learning Engineering track's Week 8 integration work now depends on the specific Week 6/7 artefacts saved for hand-off, not only the Week 4 target/feature definitions.	Machine Learning Engineering	Open — superseded in part by D12 below: the artefact to validate is now the retained 38-feature specification, not the originally-reported Week 6 model; formal hand-off still not

ID	Dependency	Track(s)	Status
			confirmed.
D7	Week 7/8 fairness and threshold-confirmation work depends on HealthConnect and/or the Project Management track supplying a formal false-positive tolerance or cost weighting.	Project Management / HealthConnect stakeholders	Open; Week 7's T07 (Week 7 report, Section 3.2) now supplies the quantified trade-off evidence this decision needs.
D8	Any future patient-grouped validation of the Week 6/7 threshold and stability work depends on patient_id being recovered or re-attached to the processed feature matrices.	Data Science (internal)	Open; still unresolved after Week 7's T01 and T06 testing, both of which used StratifiedKFold on individual appointments as the practical alternative.
D9	The Data Analytics cross-track exchange now has a concrete, auditable evidence artefact (a shared dashboard file) rather than only a narrative description.	Data Analytics	Satisfied — evidence recorded in the Week 6 report, Section 11.4.
D10	The Week 7 T08 harmonisation creates an explicit, ongoing dependency: Data Science's segment reporting must stay aligned to the Data Analytics Week 7 lead-time bands and prior-no-show-history groups, and any future change to either track's definitions will require a re-harmonisation retest.	Data Analytics	Open — ongoing; satisfied for the current Week 7 definitions (Week 7 report, Section 13).
D11	Resolving the two cross-track items raised by Data Analytics in Week 7 — prediction-level appointment-ID traceability and the missing-distance discrepancy — depends on evidence or investigation from the Data Analytics track, or from whichever track owns the source data extract.	Data Analytics	Open.
D12	The decision to retain the complete 38-feature specification rather than the	Machine Learning Engineering	Open; supersedes the

ID	Dependency	Track(s)	Status
	previously-reported 35-feature ablated model (Week 7 report, Section 8) creates a new, explicit dependency on ML Engineering to validate the correct (38-feature) artefact and schema.		artefact scope originally referenced in D6.

5. Technical limitations

- Cross-validation used for Week 6 threshold selection could not preserve patient-level grouping (Section 3, NI1), a methodological gap inherited from the absence of patient_id in the saved processed matrices.
- No automated experiment-tracking or model-registry tooling is in place; model artefacts are currently version-managed manually (.pkl and .csv files saved per week), which increases the risk of the kind of documentation/code inconsistency already encountered once (R8).
- Random Forest and Histogram Gradient Boosting have not been hyperparameter-tuned or regularised, so their true technical potential relative to Logistic Regression remains unknown (Week 6 report, Section 5.1).
- Reproducibility safeguards introduced in Week 6 (explicit assertion checks confirming reloaded artefacts match the Week 5 baseline) are a genuine strength, but formal automated testing — for example, unit tests on the feature-engineering pipeline — is not yet in place.
- Preprocessing (scaling, encoding) in the Week 7 testing notebook was performed once, prior to the fold split, rather than within a fully encapsulated per-fold pipeline, carrying a small risk of optimistic performance estimates relative to a pipeline that fits preprocessing independently inside each fold.
- As a linear model, logistic regression — even with three engineered interaction terms — cannot capture non-linear or threshold-based relationships that earlier exploratory analysis hinted at, particularly for previous_no_shows and distance_to_clinic_km.

6. Data limitations

- The HealthConnect Appointment Dataset is fictional and synthetic (R4); findings should be treated as illustrative of methodology rather than clinically validated, an assumption accepted since Week 4 and unchanged.
- Feature completeness: no socio-economic status, specific medical condition, or transportation-access data is available, which may constrain the model's predictive ceiling (Week 6 report, Section 10.1).
- The Sunday-appointment anomaly (R7) remains unresolved at the data-quality level; whether it reflects a data error or an undocumented clinic exception is still unconfirmed.
- waiting_time_minutes remains excluded from the feature set pending formal confirmation of its definition, representing a source of potentially useful signal that is currently, at minimum, deferred.

- The dataset's relatively small size (5,000 appointments; 1,696 unique patients) limits the granularity at which subgroup fairness testing (Section 9) can be performed with statistical confidence.
- The missing-distance discrepancy identified by Data Analytics during the Week 7 T08 cross-track activity (86 records with missing distance data in their source versus 39 in the Data Science working dataframe) indicates the two tracks are not working from identically filtered extracts of the same underlying data (Week 7 report, Section 13).

7. Analytical limitations

- The Week 6 comparison of Data Analytics' descriptive findings against the model's coefficients and permutation importance, and the Week 7 T08 cross-track retest (Week 7 report, Section 13), are both directional/associative evidence, not causal validation.
- Categorical variables (gender, appointment_type, reminder_sent) have still not been formally tested for association with no-show status using an appropriate independence test such as chi-square (Section 2); Week 7's T04 segment testing is descriptive rather than inferential.
- The Week 7 error-analysis feature comparisons (Week 7 report, Section 5) report standardised means by error type, not effect-size-tested group differences, so the magnitude of a mean gap should be read as descriptive rather than as a formally tested effect.

8. Modelling limitations

- Linearity: even with three engineered interaction terms, the Refined Logistic Regression may not capture more complex, non-linear relationships between features and no-show behaviour.
- Confirmed not-yet-robust improvement: the ROC-AUC/PR-AUC gain over the Week 5 baseline is small and Week 7's T02 fold-matched comparison confirms it is not robust across resamples (Section 1; Section 3, NI3) — this is now a settled finding, not an open question.
- Untuned non-linear alternatives: Random Forest and Histogram Gradient Boosting overfit substantially with default hyperparameters and cannot currently be considered as replacement candidates (Section 3, NI2).
- Threshold dependency: classification performance is highly sensitive to the chosen threshold; the provisional 0.45 threshold, now quantified by Week 7's T07 (Section 2), has still not been confirmed against a HealthConnect-defined cost structure.
- Reliance on historical, potentially non-stationary data: changes in patient demographics, booking behaviour or clinic policy could cause drift, and no production monitoring plan has yet been implemented — only proposed (Week 6 report, Section 12.5).
- The calendar-feature ablation decision rests on a narrow, pre-declared ± 0.002 reproducibility tolerance rather than a clinically or operationally derived threshold (Section 3, NI10).
- The final holdout set used for Week 7 retesting has already been evaluated in earlier weeks and is therefore not a completely unseen, external validation set, reducing its evidential strength for a final go/no-go decision (Week 7 report, Section 9).

9. AI safety/reliability concerns

The concerns below are framed with reference to the NIST AI Risk Management Framework's core functions of governing, mapping, measuring and managing AI risk throughout a system's life cycle (Tabassi, 2023), which this track adopted as a reference framework from Week 4 onward.

- Algorithmic bias and fairness: subgroup performance (e.g., by gender, age group, or distance to clinic as an access proxy) has not yet been formally tested; this remains open since Week 4 (R3) despite Week 7's T04 descriptive segment testing (Section 2), and is now more pressing because the provisional 0.45 threshold changes who is flagged for intervention — a risk well documented in healthcare algorithms generally (Obermeyer et al., 2019).
- Automation bias and over-reliance: the model is intended to support, not replace, clinical and administrative judgement, but no human-in-the-loop review protocol has yet been formally documented for triggered interventions, beyond being proposed as a mitigation.
- Patient trust and perceived surveillance: the HealthConnect-aligned 0.45 threshold increases False Positives on the holdout test set (225 versus 176 at the default threshold), meaning more attending patients would be flagged and potentially contacted unnecessarily; this trade-off has not yet been validated with HealthConnect stakeholders or assessed for its effect on patient trust.
- Data privacy and security: patient data used for training and inference must continue to be handled in line with applicable privacy and healthcare-data obligations; no separate privacy or security review has been documented to date.
- Reliability under distribution shift: no drift-detection or monitoring mechanism is yet operational, so any degradation in reliability after deployment would not currently be automatically detected.
- Reliability of the artefact-generation process itself: Week 7's own verification work (Section 1, NI8) shows that a notebook's narrative interpretation cells can silently drift from its own executed output after a retrain, with no automatic signal that they have gone stale. This is a process-level AI-reliability risk worth carrying into Week 8 practice — for example, regenerating all narrative cells immediately after any retrain, rather than relying on a later manual verification pass to catch drift.

10. Integration challenges

- Cross-track synchronisation remains incomplete: the Data Analytics exchange now has two documented, auditable evidence artefacts (the Week 6 dashboard, D9, and the Week 7 T08 harmonisation, D10), but the Machine Learning Engineering, Project Management and Generative AI syncs proposed in the Week 5 Track Coordination Plan have not been confirmed as completed.
- No formally agreed hand-off format with the Machine Learning Engineering track: the Week 6/7 artefacts (model file, refined feature CSVs, frozen threshold) were saved locally, but no integration contract — for example, expected input schema or inference-latency requirements — has yet been documented jointly with that track, and the artefact itself has changed (Section 4, D12: the retained 38-feature specification, not the previously-reported 35-feature ablated one).

- The Generative AI track's Clinic Knowledge Base states the clinic is closed on Sundays, which conflicts with the unresolved Sunday-appointment anomaly (R7); until this is escalated and resolved, the two tracks are effectively working from inconsistent assumptions about clinic operating hours.
- Documentation debt (the README's uncorrected split-method description) creates a risk that a track relying on the notebook's narrative documentation, rather than its executed code, could work from an inaccurate description of the modelling methodology.
- The two Data Analytics cross-track items raised in Week 7 (appointment-ID traceability, missing-distance discrepancy) depend on evidence or timelines outside this track's direct control (Section 2; Section 4, D11).

11. Testing limitations

- Without patient-grouped cross-validation (Section 5), Week 7's T01 stability and T06 generalisation findings may not fully capture repeat-patient generalisation risk.
- T04's segment screening used a practical $n \geq 50$ cutoff to avoid reporting unstable small-sample rates (Week 7 report, Section 6); very small segments' true error rates are therefore not reported and could hide localised risk.
- T01 and T02 rely on descriptive variability across five folds rather than formal statistical significance testing (for example, confidence intervals or paired significance tests on fold differences); conclusions about stability and the baseline comparison should be read with this in mind.
- Most Week 7 tests (T01–T06, T08) were conducted via cross-validation on the training partition, with the held-out test set used only once for final confirmation (Week 7 report, Section 9); that held-out set is itself reused from earlier weeks rather than freshly unseen (Section 8).
- Random Forest and Histogram Gradient Boosting have still not been hyperparameter-tuned (Section 3, NI2), so Week 7's testing cycle was conducted entirely on the Logistic Regression family and says nothing about whether a properly tuned non-linear alternative would test differently.

12. Potential impact on the project

- If subgroup fairness is not tested before any pilot or deployment decision, HealthConnect risks a model that under-serves or over-targets specific patient groups, with reputational and ethical consequences for the wider Experience Lab project.
- If the provisional 0.45 threshold is adopted operationally without stakeholder confirmation, HealthConnect could face an intervention burden (225 unnecessary contacts observed on the holdout set) that exceeds actual outreach capacity, or could otherwise retain the default threshold's higher missed-no-show rate unnecessarily.
- Continued delay in cross-track synchronisation (Machine Learning Engineering, Project Management, Generative AI) increases the risk that downstream integration work is designed against outdated assumptions rather than the current, Week 7-retained 38-feature model artefact.

- Unresolved data-quality questions (the Sunday-appointment anomaly, `waiting_time_minutes`, and the Week 7 missing-distance discrepancy) leave residual uncertainty in the underlying dataset that could affect confidence in any headline result presented at the Week 8 Final Presentation.
- The Week 7 testing cycle has now confirmed, rather than merely risked, that the Week 6 performance gain over the baseline is not robust (Section 1); if this is not communicated clearly to stakeholders, HealthConnect could over-invest confidence in the engineered-feature model.
- Unresolved appointment-ID traceability (Section 2) could block HealthConnect's ability to audit specific flagged patients ahead of a production rollout, potentially delaying Week 8 sign-off.

13. Recommended mitigation

The following mitigations are recommended, prioritised by the severity of potential impact set out in Section 12:

Priority	Issue / Risk	Recommended Mitigation	Owner	Target
High	Subgroup fairness untested (R3)	Conduct formal fairness testing (e.g., demographic parity, equal opportunity, predictive parity) across gender, age group and distance-based access proxy before any deployment discussion; Week 7's T04 descriptive segment testing is not a substitute.	Data Science	Week 8
High	Provisional threshold not confirmed (NI4)	Present the Week 7 T07 threshold-trade-off evidence (Week 7 report, Section 3.2) to HealthConnect stakeholders and obtain a formal false-positive tolerance or cost weighting.	Data Science / Project Management	Week 8
High	Sunday-appointment anomaly unescalated (R7)	Formally escalate to the data owner or mentor for confirmation, coordinating with the Generative AI track given the shared Clinic Knowledge Base dependency.	Data Science / Project Management	Week 8 (overdue from Weeks 6–7)
High	Two Data Analytics cross-track items open (Section 2)	Resolve prediction-level appointment-ID traceability and the missing-distance discrepancy (86 vs. 39 records) jointly with the Data Analytics track.	Data Science / Data Analytics	Week 8
High	ML Engineering pipeline validation not completed (D6, D12)	Schedule ML Engineering pipeline validation as an explicit, tracked task targeting the retained 38-feature artefact, feature schema and threshold — not the previously-reported 35-feature ablated specification.	Data Science / Machine Learning Engineering	Week 8

Priority	Issue / Risk	Recommended Mitigation	Owner	Target
Medium	Cross-track syncs incomplete (R6)	Schedule and document formal syncs with Machine Learning Engineering, Project Management and Generative AI, each with a recorded evidence artefact, as achieved with Data Analytics (D9, D10).	Data Science / Project Management	Week 8
Medium	Patient identifiers absent from processed matrices (NI1, D8)	Recover or re-attach patient_id to the processed feature matrices (without using it as a model feature) specifically to enable patient-grouped cross-validation, still unresolved after Week 7's T01/T06.	Data Science	Week 8
Medium	Untuned tree-based models (NI2)	Perform systematic hyperparameter tuning and regularisation of Random Forest and Histogram Gradient Boosting before drawing conclusions about model-family suitability.	Data Science	Week 8 (if scope allows)
Medium	README/documentation inconsistency (R8)	Correct the README's description of the train/test split method to match the patient-aware method actually used.	Data Science	Week 8
Medium	Narrative/executed-output staleness recurs across weeks (NI8)	Adopt a standing practice of regenerating every notebook narrative/interpretation cell immediately after any retrain, rather than relying on a later verification pass to catch drift.	Data Science	Week 8 onward
Medium	Appointment-type label mismatch in T04 narrative (NI9)	Correct the "New Patient" label mismatch in the T04 narrative commentary cell so it matches the segment's four actual executed groups.	Data Science	Week 8
Low	Ablation tolerance sensitivity (NI10)	Treat the ± 0.002 tolerance as provisional and revisit it with clinical/stakeholder input if the model is retrained in a future week.	Data Science	Week 8/9
Low	waiting_time_minutes formal confirmation (R1)	Obtain written confirmation from the data owner of the field's definition, even though the feature is already excluded, to close the item formally.	Data Science	Week 8
Low	Cancellation-exclusion definition (R2)	Confirm with HealthConnect stakeholders whether excluding Cancelled appointments matches operational priorities.	Data Science / Project Management	Week 8

Consistent with the review cadence established in the Week 4 Risk and Dependency Register, this consolidated report should be reviewed and updated at the start of each subsequent week of the Experience Lab, with High-priority items escalated immediately rather than held for the next scheduled review (PMI, 2021).

14. Issues that must be addressed before Week 8

Drawing on the High-priority rows of Section 13, the following items specifically gate Week 8 sign-off and should not be deferred further:

1. Resolve the two outstanding Data Analytics cross-track items: prediction-level appointment-ID traceability and the missing-distance discrepancy (Section 2).
2. Obtain a formal HealthConnect stakeholder decision on the 0.45 threshold's recall/false-positive trade-off, now that Week 7's T07 has quantified it precisely (Section 2).
3. Complete ML Engineering pipeline validation of the final (38-feature) model artefact, feature schema and threshold (Section 4, D12).
4. Escalate the Sunday-appointment data-quality anomaly to the data owner, since it continues to be used as a live model feature while its cause remains unconfirmed and this action is now overdue from Weeks 6–7 (Section 2).
5. Conduct formal subgroup fairness testing before any deployment discussion, since Week 7's T04 descriptive segment testing does not substitute for it (Section 9).
6. Prepare concise Week 8 presentation evidence that follows the Week 6 output → Week 7 test → finding → refinement → retest → readiness-decision narrative, explicitly noting the ablation-decision reversal so the testing process itself, not only its final metrics, is visible to reviewers.

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